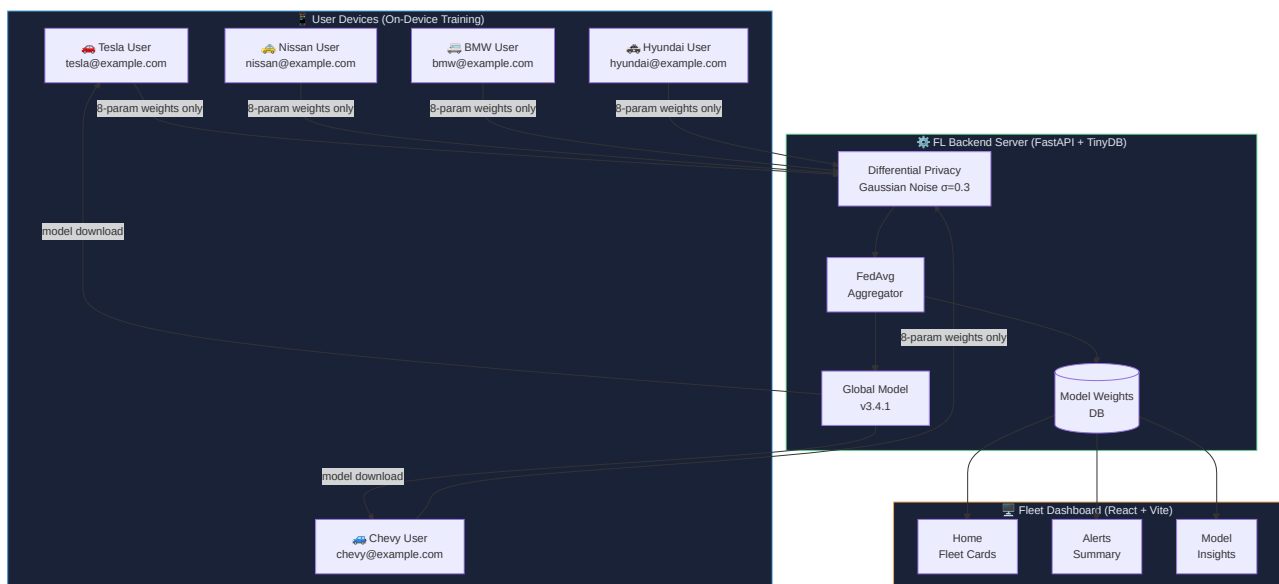


Federated Learning for EV Fleet Management

A privacy-preserving AI system that predicts electric vehicle battery degradation by training machine learning models **directly on users' phones** — without ever sending raw personal data to a server.

System Overview



Tech Stack

Layer	Technologies
Mobile App	React Native, Expo, TypeScript, Axios
On-Device ML	Custom Gradient Descent (JavaScript) — no TensorFlow
Mobile Storage	Expo AsyncStorage (model weights only)
FL Backend	Python FastAPI, TinyDB, bcrypt, JWT
FL Algorithm	FedAvg + Gaussian-noise Differential Privacy
Fleet Dashboard	Vite, React, Tailwind CSS, Recharts
Salesman Backend	Node.js, Express
Training Data	Real EV charging patterns CSV (244 Tesla samples) + synthetic

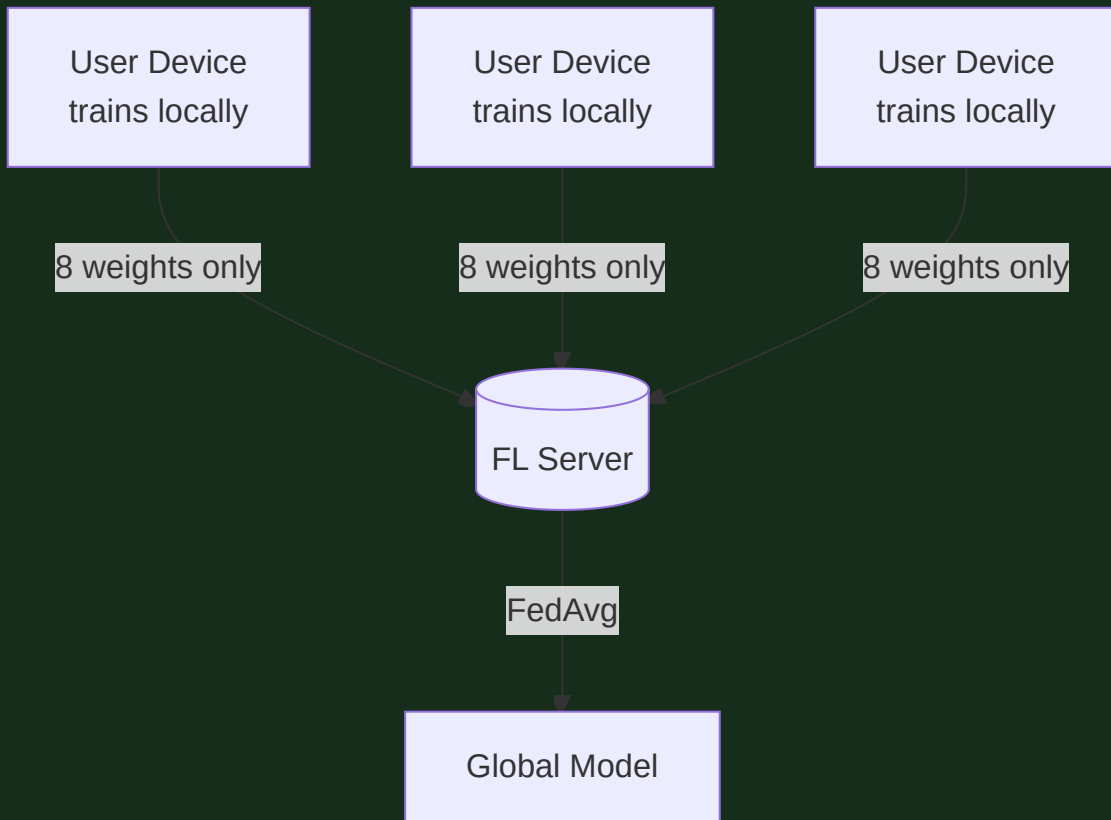
01 — The Problem

Electric vehicle owners generate rich telemetry — driving patterns, charging behavior, battery metrics. Fleet managers need this data to predict failures and plan maintenance. But collecting raw driving data raises serious **privacy and legal concerns**.

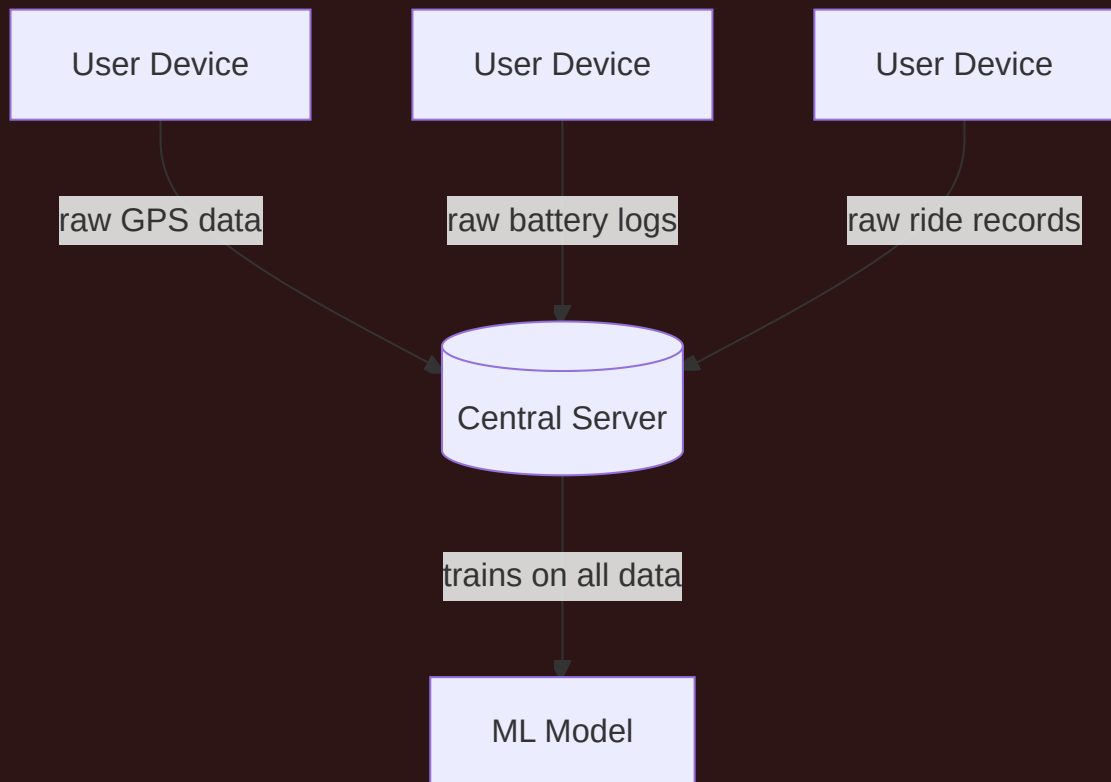
This system takes the opposite approach: the AI goes to the data, not the other way around.

Traditional ML vs Federated Learning

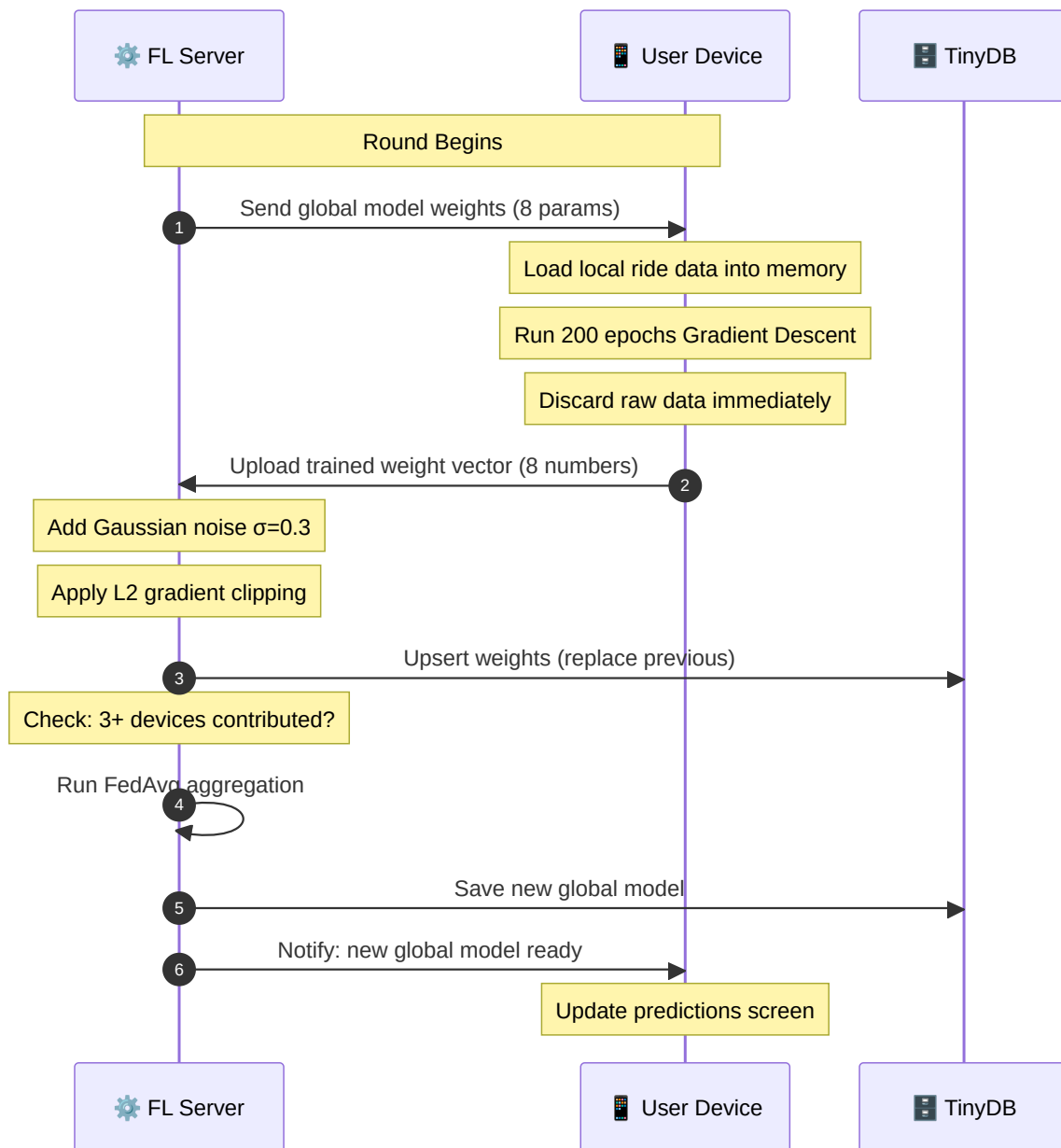
✓ Federated Approach



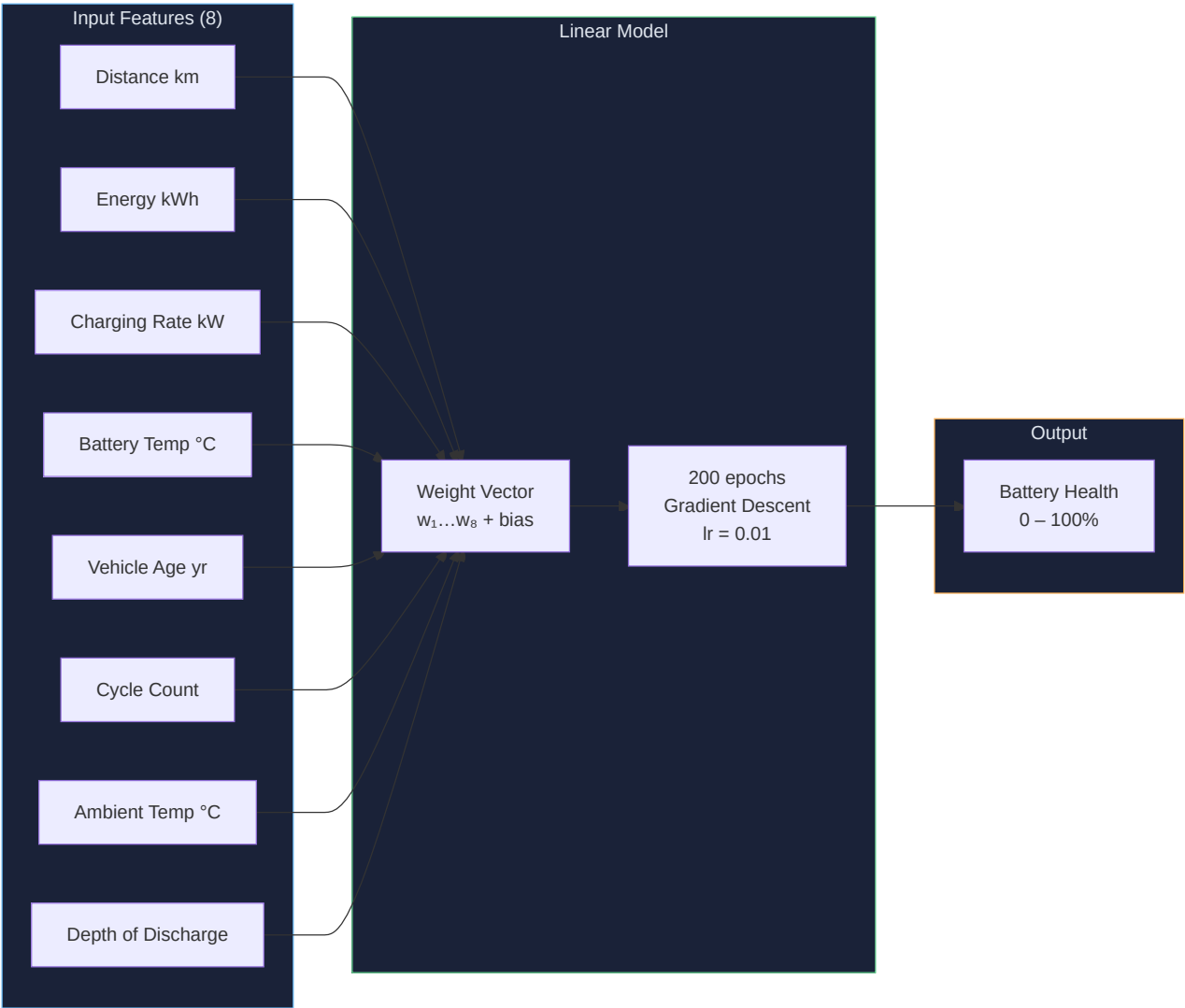
✗ Traditional Approach



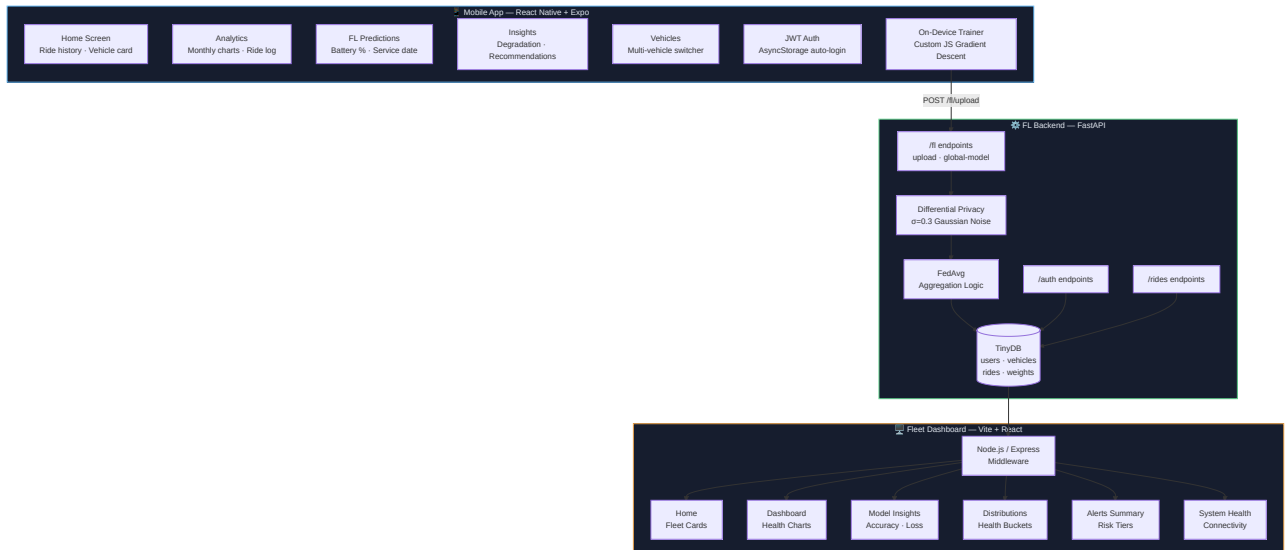
02 — Federated Learning Cycle



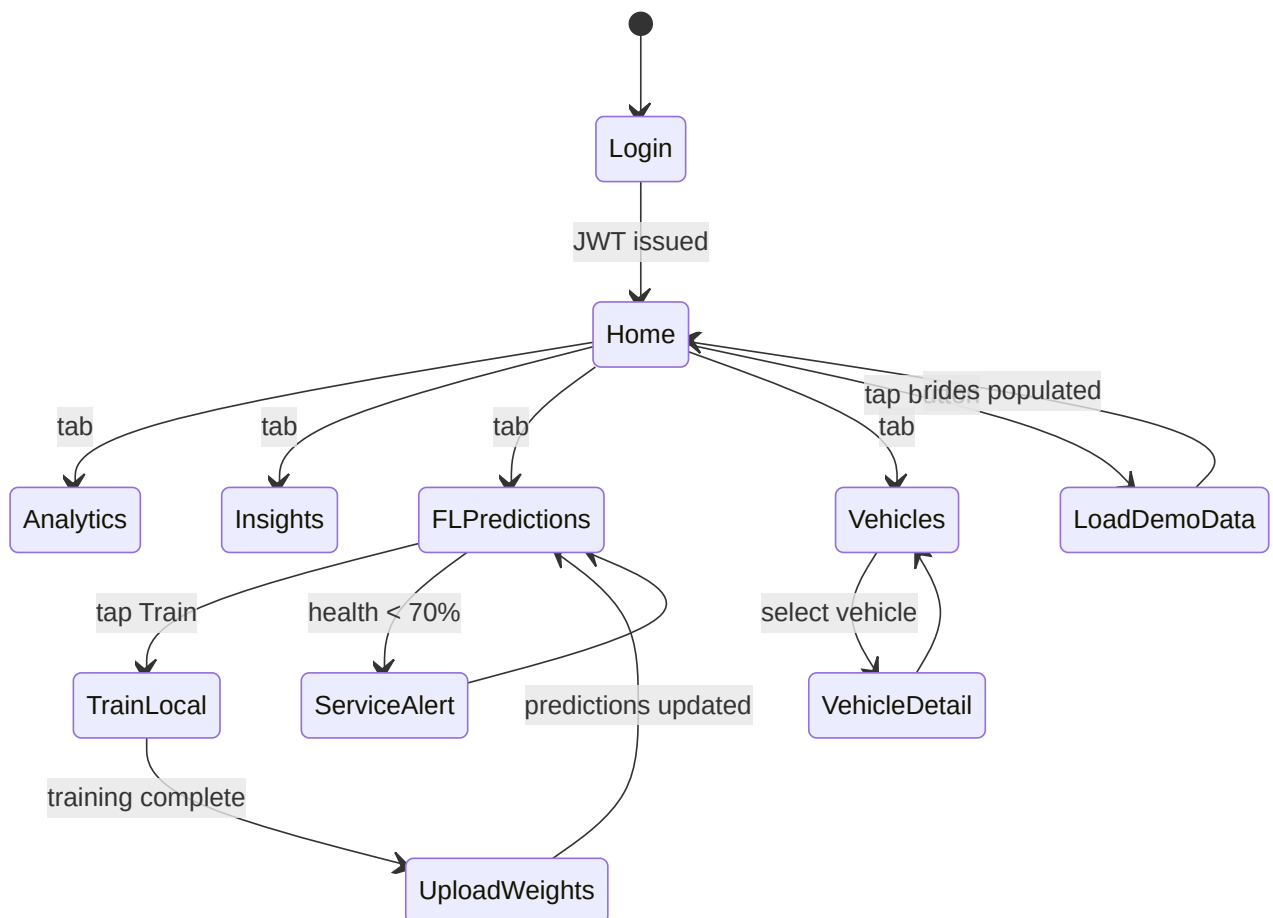
Training Details



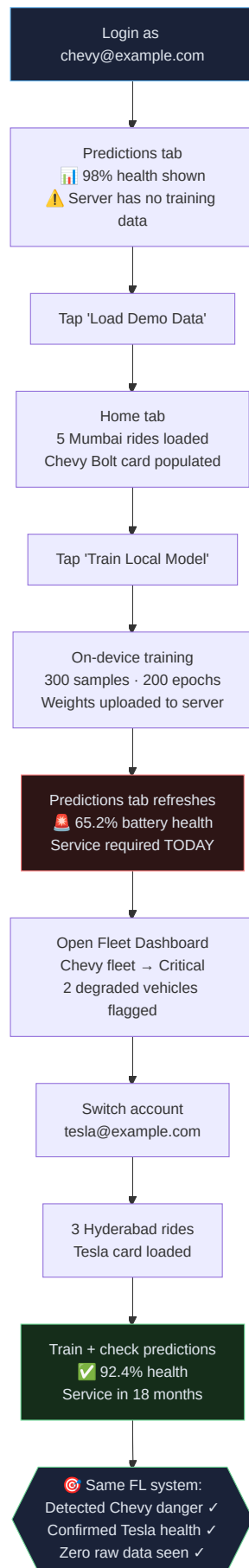
03 — System Architecture



Mobile App — Screen Flow



04 — The Demo: “Before and After”



Demo Step-by-Step Table

Step	Login / Action	Screen	Result
1	<code>chevy@example.com</code> login	Predictions tab	98% health — server clueless
2	Tap “Load Demo Data”	Home tab	5 Mumbai rides · Chevy Bolt card
3	Tap “Train Local Model”	Progress indicator	300 samples trained · weights uploaded
4	Predictions refresh	FL Predictions	65.2% 🚨 Service TODAY
5	Open fleet dashboard	Alerts Summary	Chevy → Critical · 2 degraded
6	<code>tesla@example.com</code> login	Home tab	3 Hyderabad rides · Tesla card
7	Train + check predictions	FL Predictions	92.4% ✅ Service in 18 months

05 — Fleet Status

 Critical <70%

Chevy Bolt
chevy@example.com
Mumbai
65.2%

Nissan Leaf
nissan@example.com
Chennai
63.8%

Multi Vehicle
multi@example.com
Mumbai+Chennai
2 critical

 Moderate 80–90%

BMW i3
bmw@example.com
New Delhi
84.6%

 Stable >90%

Tesla Model 3
tesla@example.com
Hyderabad
91.8%

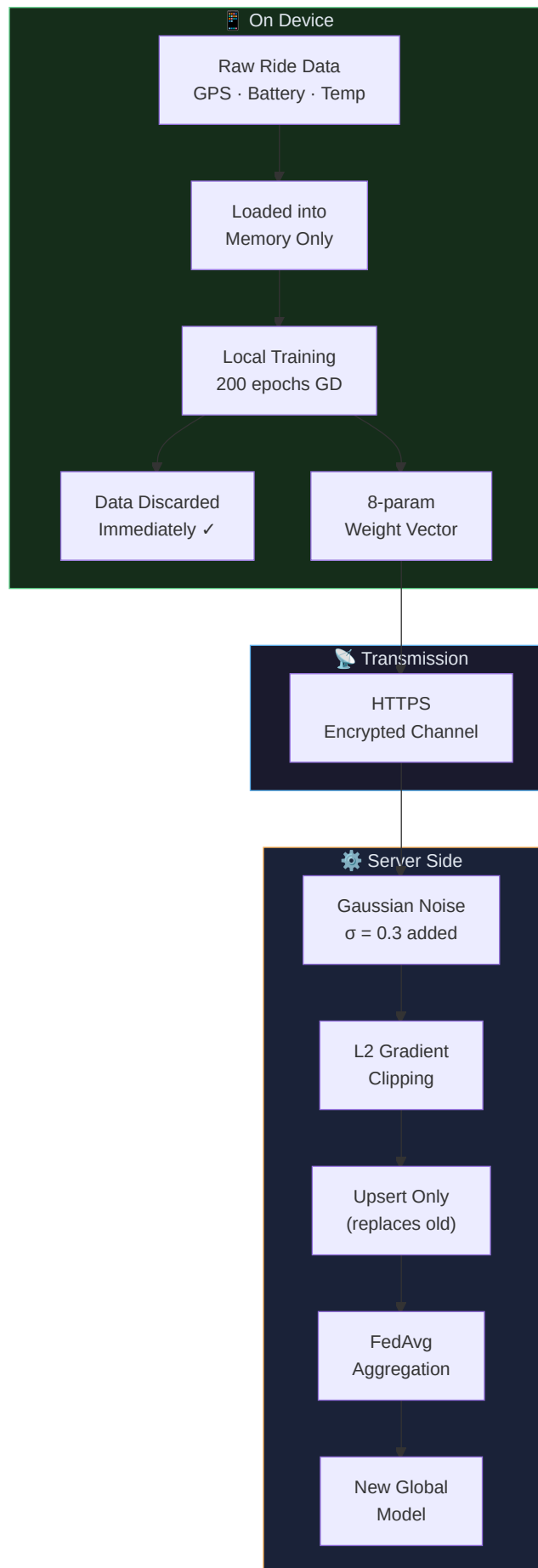
Hyundai Kona
hyundai@example.com
Bengaluru
93.2%

All Credentials

Email	Vehicle	City	Health	Status
tesla@example.com	Tesla Model 3	Hyderabad	91.8%	✔ Stable
hyundai@example.com	Hyundai Kona	Bengaluru	93.2%	✔ Stable
bmw@example.com	BMW i3	New Delhi	84.6%	⚠ Moderate
chevy@example.com	Chevy Bolt	Mumbai	65.2%	🚨 Critical
nissan@example.com	Nissan Leaf	Chennai	63.8%	🚨 Critical
multi@example.com	Chevy Bolt + Nissan Leaf	Mumbai / Chennai	—	🚨 2 Critical

All passwords: demo123

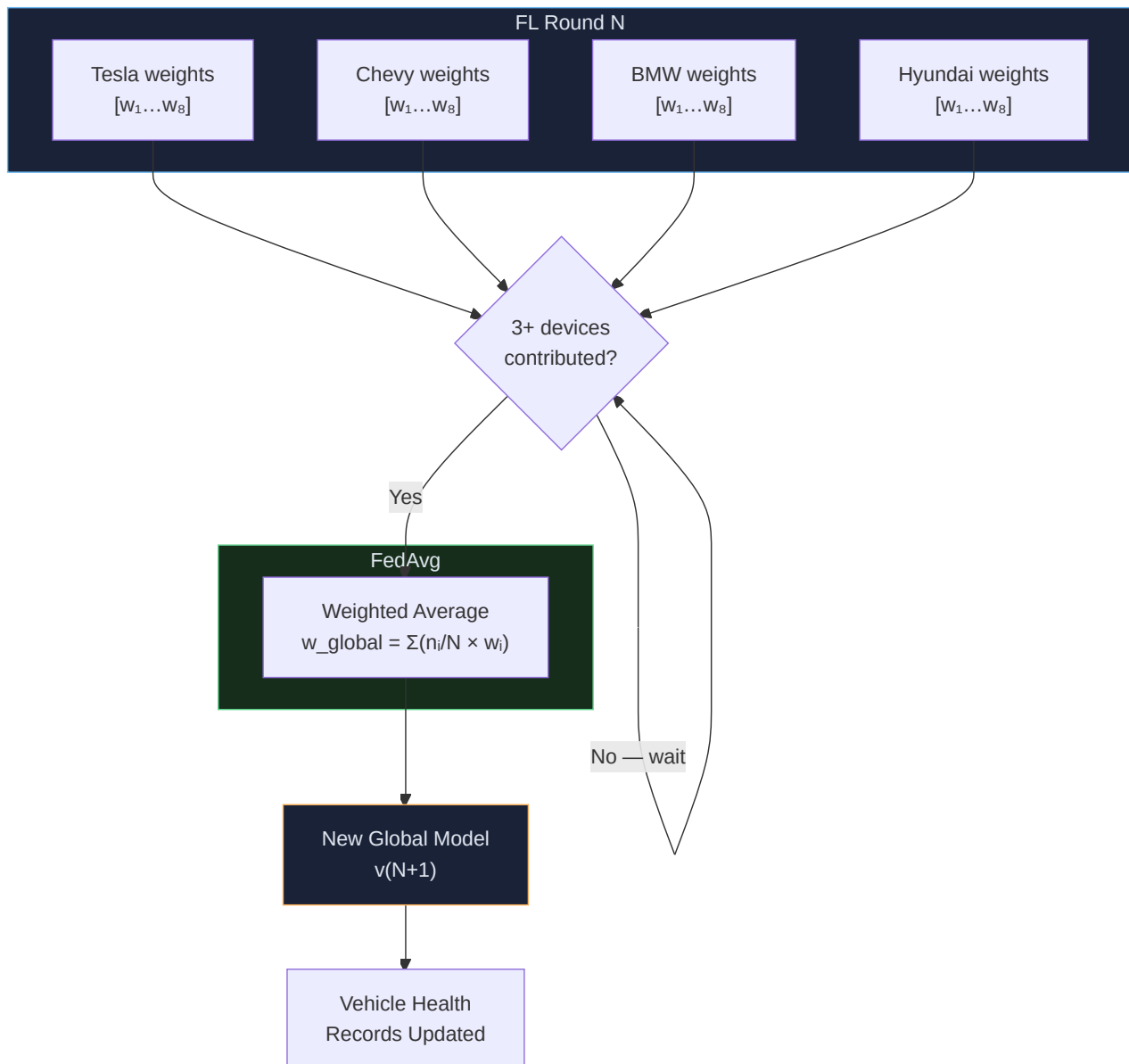
06 — Privacy Architecture



Privacy Guarantees

Guarantee	Mechanism	Detail
✓ Raw data never leaves device	Memory-only loading	Loaded → trained → discarded; never hits disk
✓ Only weights transmitted	8-number vector	Mathematically impossible to reconstruct original data
✓ Differential Privacy	Gaussian noise $\sigma=0.3$	Added server-side before any storage
✓ Gradient protection	L2 clipping	Prevents gradient inversion attacks
✓ No identity accumulation	Upsert policy	Retrain replaces old upload — no growing history
✓ Fleet isolation	Per-type models	Cross-fleet contamination impossible

07 — FedAvg Aggregation

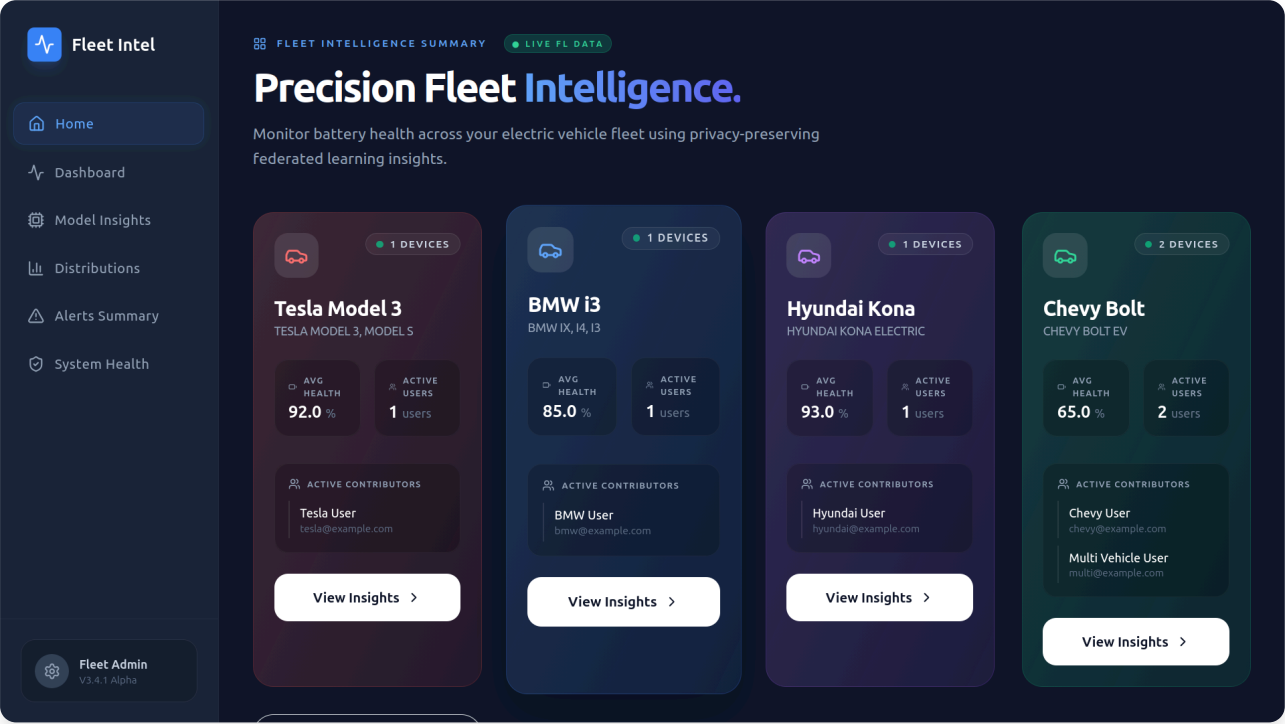


Model parameters:

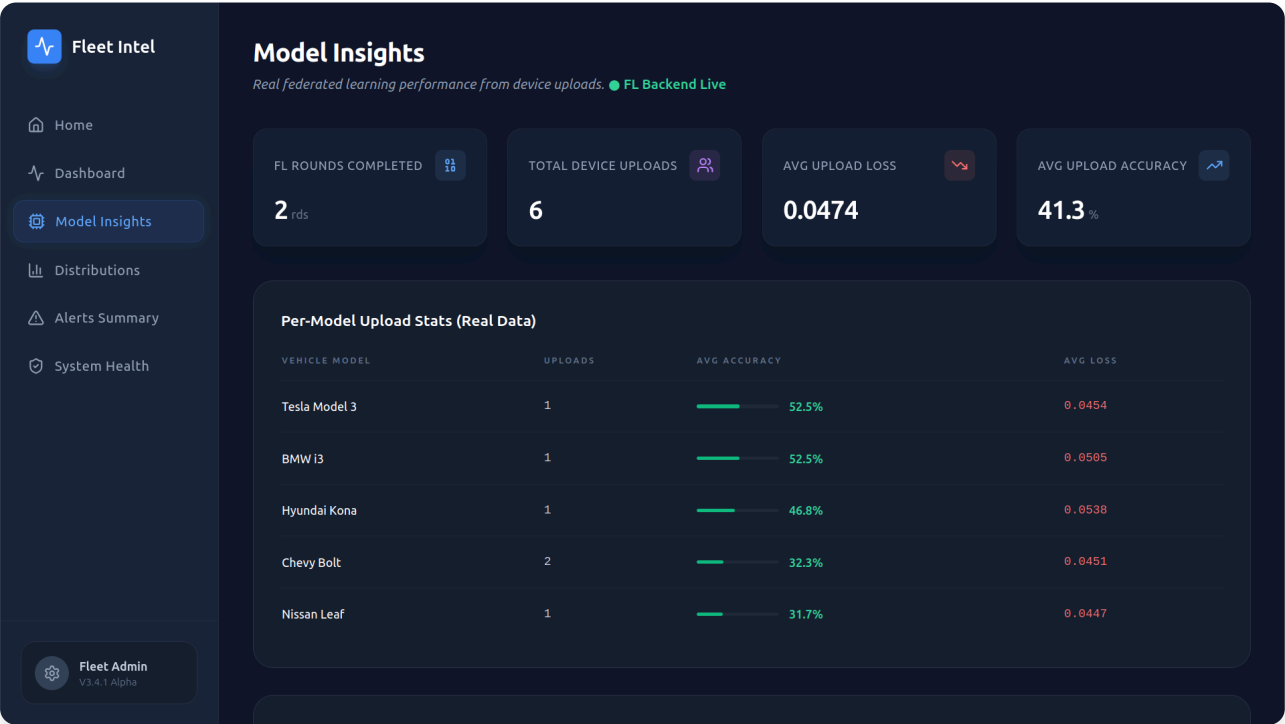
- Architecture: Linear regression, 8 learnable weights + bias
- Training: 200 epochs, learning rate 0.01, Gradient Descent
- Aggregation: FedAvg — weighted average proportional to sample count
- Personalisation: 70% global weights + 30% local training contribution

08 — Fleet Dashboard Screenshots

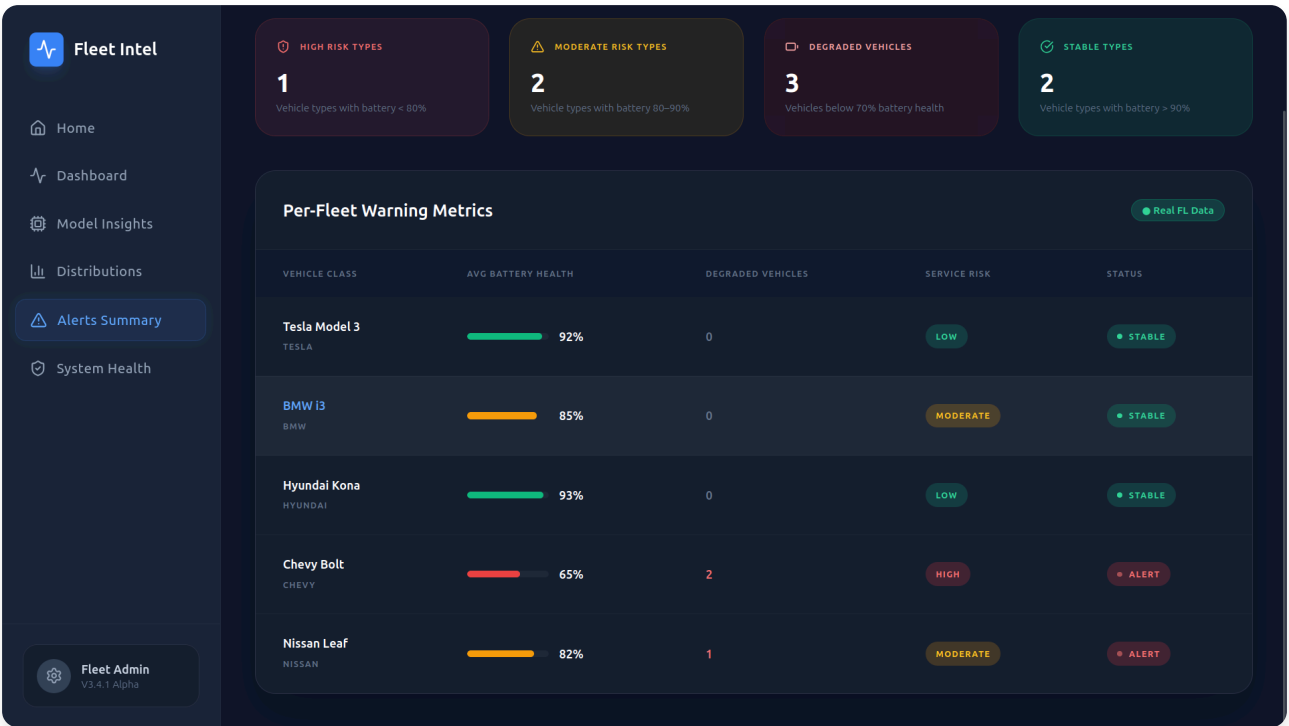
Home — Fleet Intelligence Summary



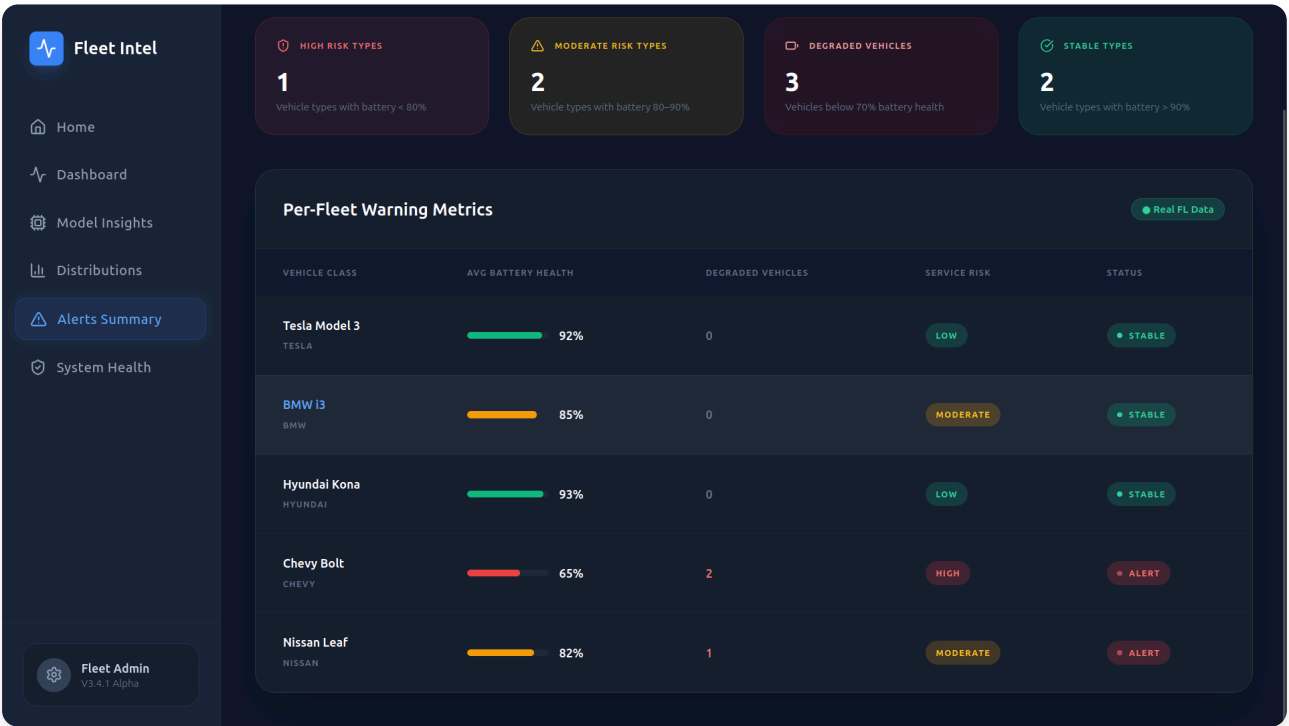
Dashboard — Fleet Overview & Battery Distribution



Alerts Summary — Per-Fleet Warning Metrics

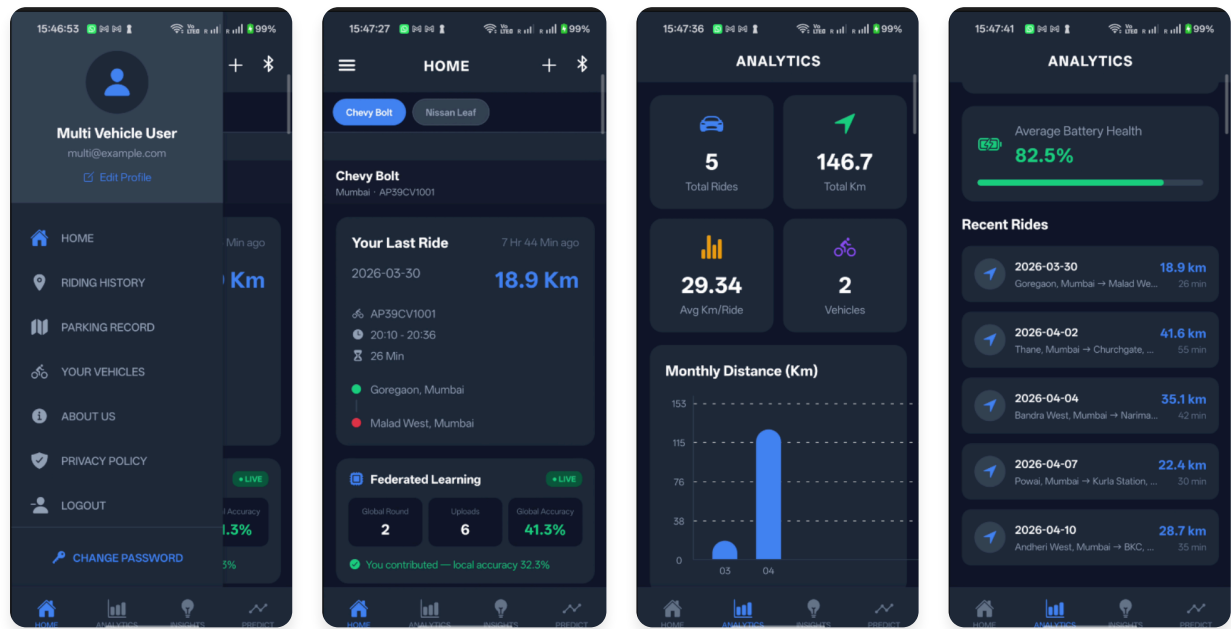


System Health — Privacy Architecture

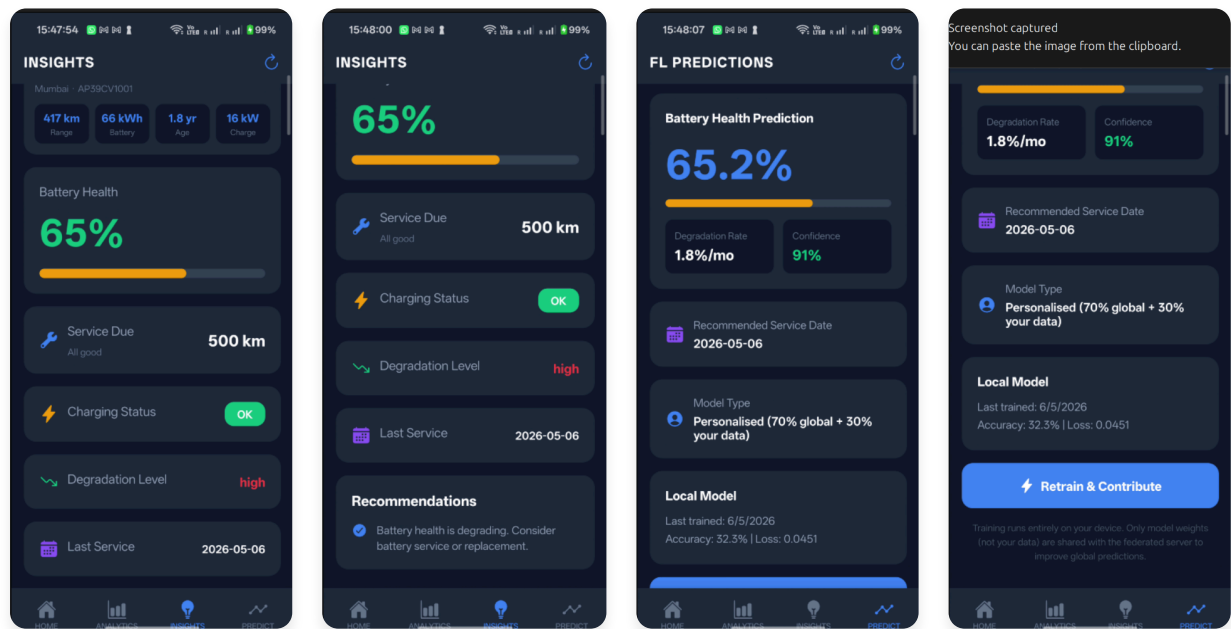


09 — Mobile App Screenshots

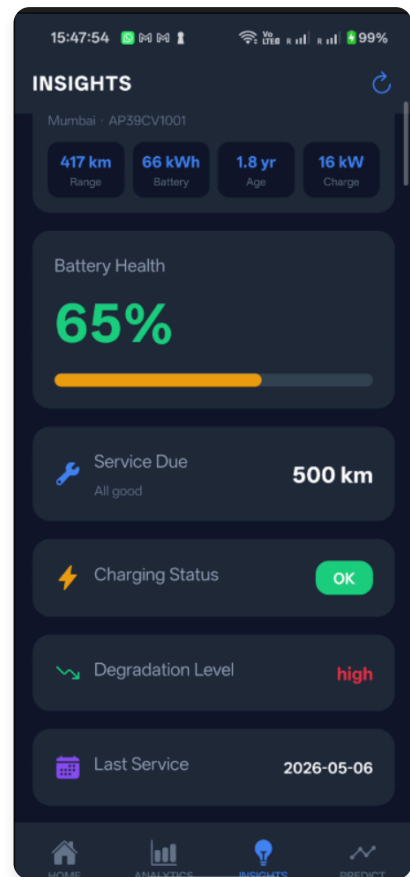
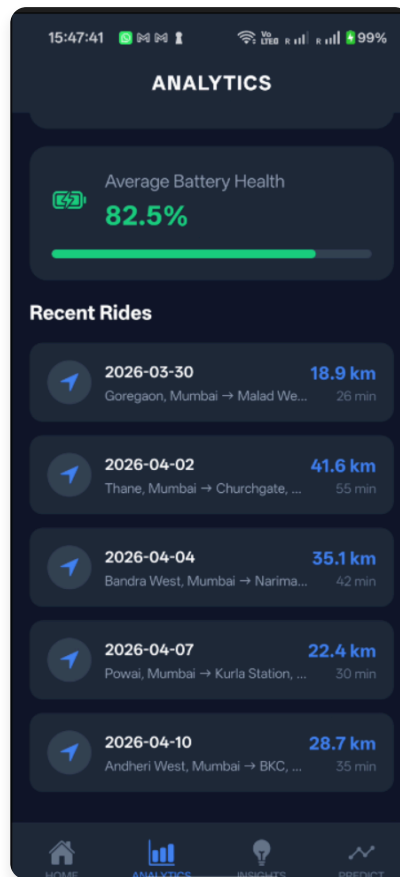
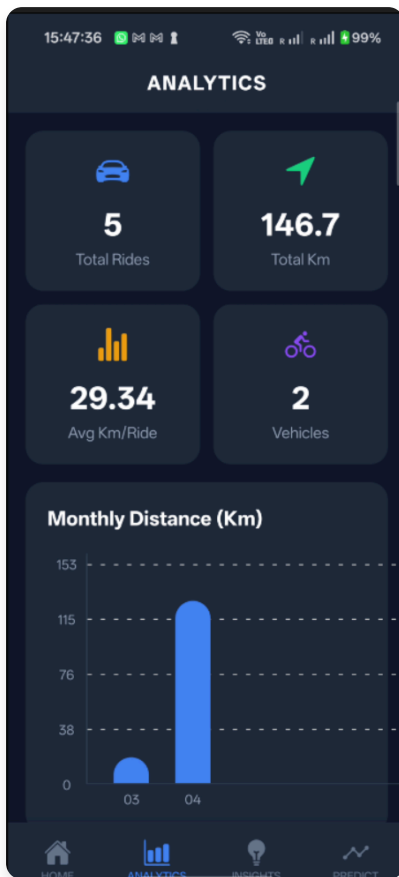
Side Menu · Home · Analytics Overview · Ride List



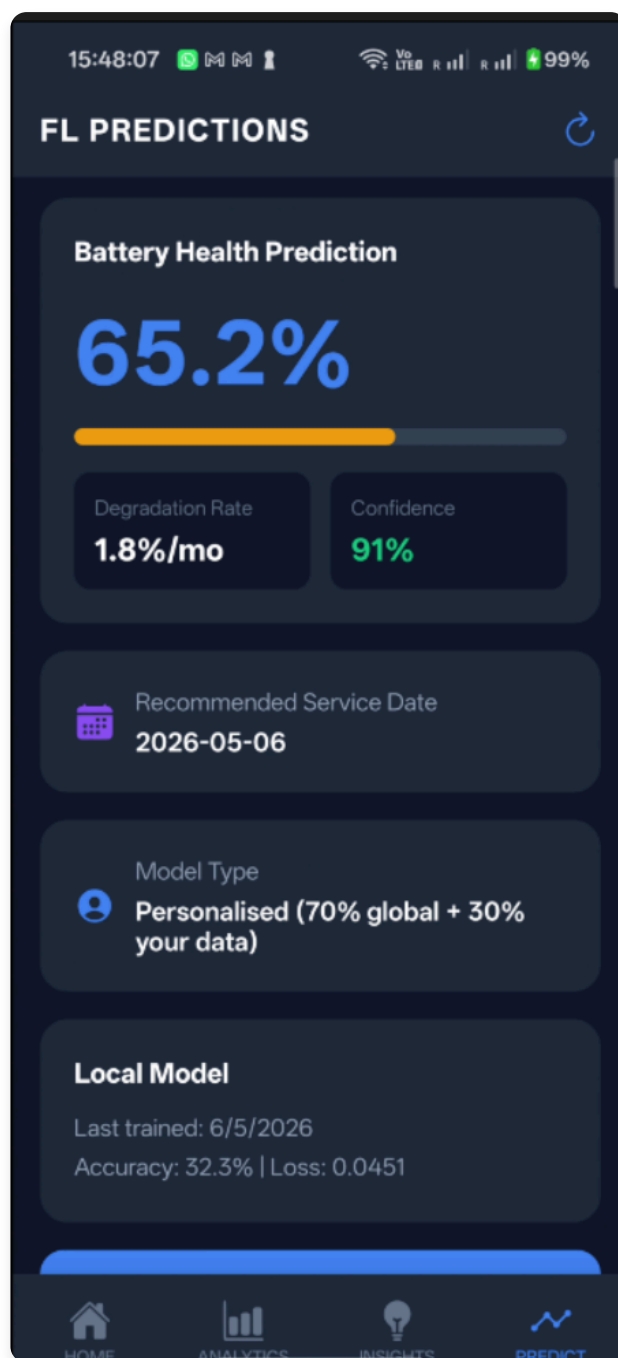
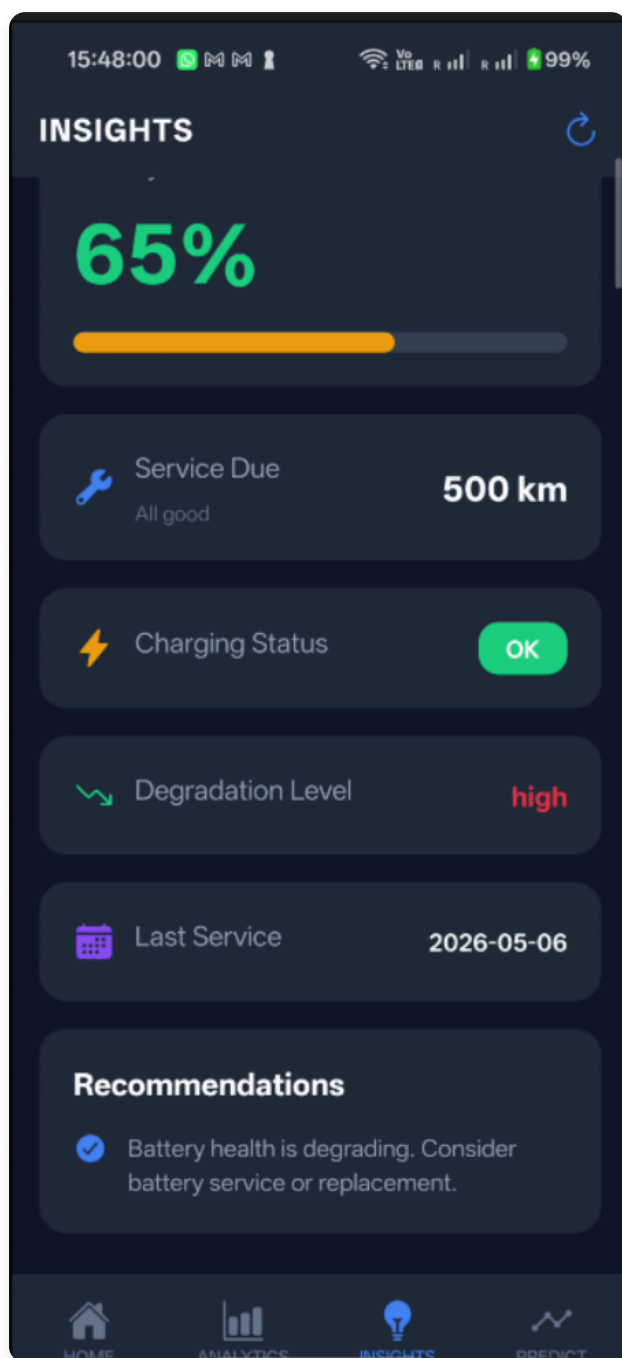
Insights · Recommendations · FL Predictions · Retrain



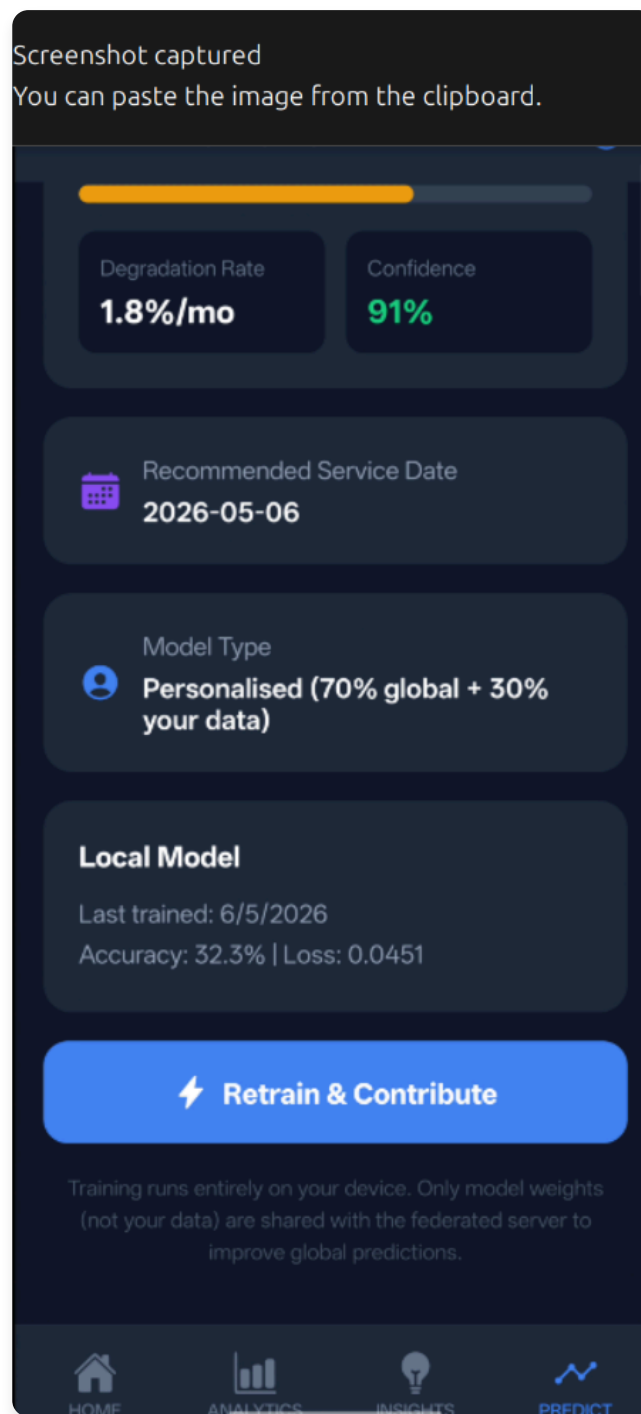
Model Insights & Predictions



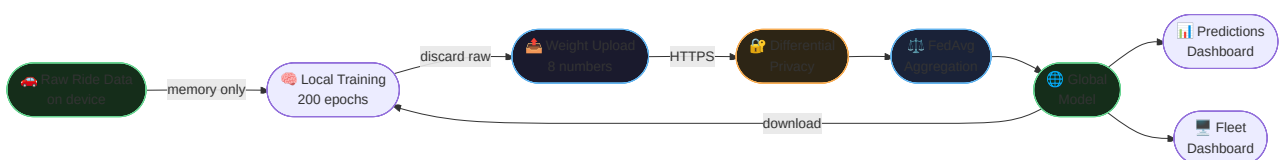
Additional Analysis



Final Summary



10 — Data Flow Summary



Summary

Component	Role	Privacy Contribution
 Mobile App	Trains local model, displays predictions	Raw data stays on-device
 FL Backend	Aggregates weights, runs FedAvg	Never stores raw data; adds noise
 Fleet Dashboard	Shows fleet health from FL outputs	Sees only aggregated metrics

One sentence: A complete, working federated learning system that detects real battery degradation across a vehicle fleet — with zero raw data ever leaving any user’s phone.